

## Project Initialization and Planning Phase

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| <b>Date</b>          | 30 September 2026                                  |
| <b>Team ID</b>       | 6ab39579211fa4fd5d28901b                           |
| <b>Project Title</b> | EduGenie: Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                            |

### Project Proposal (Proposed Solution) report

EduGenie is a Google Gemini powered learning assistant that brings common college learning activities into one web interface. It supports question

answering, concept explanation, quiz generation, summarization, and personalized learning paths.

| Project Overview    |                                                                                                                                                                                                                                             |
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| <b>Objective</b>    | Provide college students with a single, simple learning assistant that gives focused explanations, practice support, summaries and study guidance.                                                                                          |
| <b>Scope</b>        | Support five learning activities through a modular FastAPI application, using Gemini 3.5 Flash-Lite for generative features and LaMini-Flan-T5-783M for local concept explanation.                                                          |
| Problem Statement   |                                                                                                                                                                                                                                             |
| <b>Description</b>  | Students often use multiple scattered resources to understand technical concepts, practise questions, summarize content and decide what to study next.                                                                                      |
| <b>Impact</b>       | A unified learning assistant can reduce resource switching and provide focused AI-assisted support for understanding, practice, revision and study planning.                                                                                |
| Proposed Solution   |                                                                                                                                                                                                                                             |
| <b>Approach</b>     | Route each student request to the appropriate EduGenie feature through FastAPI and return a structured learning result through the responsive web interface.                                                                                |
| <b>Key Features</b> | <ul style="list-style-type: none"> <li>• Q&amp;A; using Gemini 3.5 Flash-Lite.</li> <li>• Concept explanation using LaMini-Flan-T5-783M.</li> <li>• Quiz generation, summarization and personalized learning paths using Gemini.</li> </ul> |

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|  | <ul style="list-style-type: none"> <li>• Structured learning outputs with input validation.</li> <li>• Local execution using FastAPI and Uvicorn.</li> <li>• Modular Python feature modules for maintainability.</li> </ul> |
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### Resource Requirements

| Resource Type           | Description            | Specification/Allocation                                                                                             |
|-------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>         |                        |                                                                                                                      |
| Computing Resources     | CPU / GPU              | Standard laptop/desktop CPU; GPU not required for the current MVP.                                                   |
| Memory                  | RAM                    | 8 GB RAM or higher recommended for local development.                                                                |
| Storage                 | Disk space             | At least 5 GB free space for Python packages and the local model cache.                                              |
| <b>Software</b>         |                        |                                                                                                                      |
| Frameworks              | Python web framework   | FastAPI and Uvicorn                                                                                                  |
| Libraries               | AI and Python packages | google-genai, transformers, torch, python-dotenv                                                                     |
| Development Environment | IDE / editor           | VS Code or another Python-capable IDE; browser for testing.                                                          |
| <b>Data</b>             |                        |                                                                                                                      |
| Data                    | Data requirement       | No external dataset is required for the current MVP. User-provided learning content is processed during feature use. |